
MUHAMMAD TAHA

22F-3277

BSC-3C

17/11/2023

LAB 12

(Total Marks 10)

Task 1: Write a program that hooks an interrupt zero to display your name on screen.

```
[org 0x0100]
```

```
jmp start
```

```
message: db 'Muhammad Taha', 0
```

```
clrscr:
```

```
    push es
    push ax
    push cx
    push di
    mov ax, 0xb800
    mov es, ax ; point es to video base
    xor di, di ; point di to top left column
    mov ax, 0x0720 ; space char in normal attribute
    mov cx, 2000 ; number of screen locations
    cld ; auto increment mode
    rep stosw ; clear the whole screen
    pop di
    pop cx
    pop ax
    pop es
    ret
```

```
strlen:
```

```
    push bp
    mov bp, sp
    push es
    push cx
    push di
    les di, [bp+4] ; point es:di to string
    mov cx, 0xffff ; load maximum number in cx
    xor al, al ; load a zero in al
    repne scasb ; find zero in the string
    mov ax, 0xffff ; load maximum number in ax
    sub ax, cx ; find change in cx
    dec ax ; exclude null from length
    pop di
```

```
pop cx
pop es
pop bp
ret 4
```

printstr:

```
push bp
mov bp, sp
push es
push ax
push cx
push si
push di
push ds ; push segment of string
```

```
mov ax, [bp+4]
push ax ; push offset of string
```

```
call strlen ; calculate string length
```

```
cmp ax, 0 ; is the string empty
jz exit ; no printing if string is empty
```

```
mov cx, ax ; save length in cx
```

```
mov ax, 0xb800
mov es, ax ; point es to video base
mov al, 80 ; load al with columns per row
mul byte [bp+8] ; multiply with y position
add ax, [bp+10] ; add x position
shl ax, 1 ; turn into byte offset
mov di, ax ; point di to required location
mov si, [bp+4] ; point si to string
mov ah, [bp+6] ; load attribute in ah
cld ; auto increment mode
```

nextchar:

```
lodsb ; load next char in al
stosw ; print char/attribute pair
loop nextchar ; repeat for the whole string
```

exit:

```
pop di
pop si
pop cx
pop ax
pop es
pop bp
ret 8
```

; divide by zero interrupt handler

myisrfor7:

```
push ax ; push all regs
push bx
push cx
push dx
push si
push di
push bp
push ds
push es
```

```
push cs
pop ds ; point ds to our data segment
```

```
call clrscr ; clear the screen
mov ax, 15
push ax ; push x position
mov ax, 10
push ax ; push y position
mov ax, 0x01 ; white on blue attribute
push ax ; push attribute
mov ax, message
push ax ; push offset of message
call printstr ; print message
```

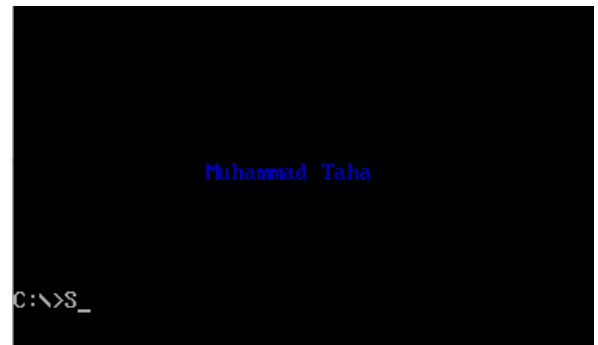
```
pop es
pop ds
pop bp
pop di
pop si
pop dx
pop cx
pop bx
pop ax
iret
```

start:

```
xor ax, ax
mov es, ax ; load zero in es
mov word [es:7*4], myisrfor7 ; store offset at n*4
mov [es:7*4+2], cs ; store segment at n*4+2
```

```
int 0x7
```

```
mov ax, 0x4c00 ; terminate program
int 0x21
```



Task 2: Execute and understand program 8.2. (viva-based marks only in lab)

; print string using bios service

[org 0x0100]

jmp start

message: db 'Taha'

start:

mov ah, 0x13 ; service 13 - print string

mov al, 1 ; subservice 01 – update cursor

mov bh, 0 ; output on page 0

mov bl, 7 ; normal attrib

mov dx, 0x0A03 ; row 10 column 3

mov cx, 4 ; length of string

push cs

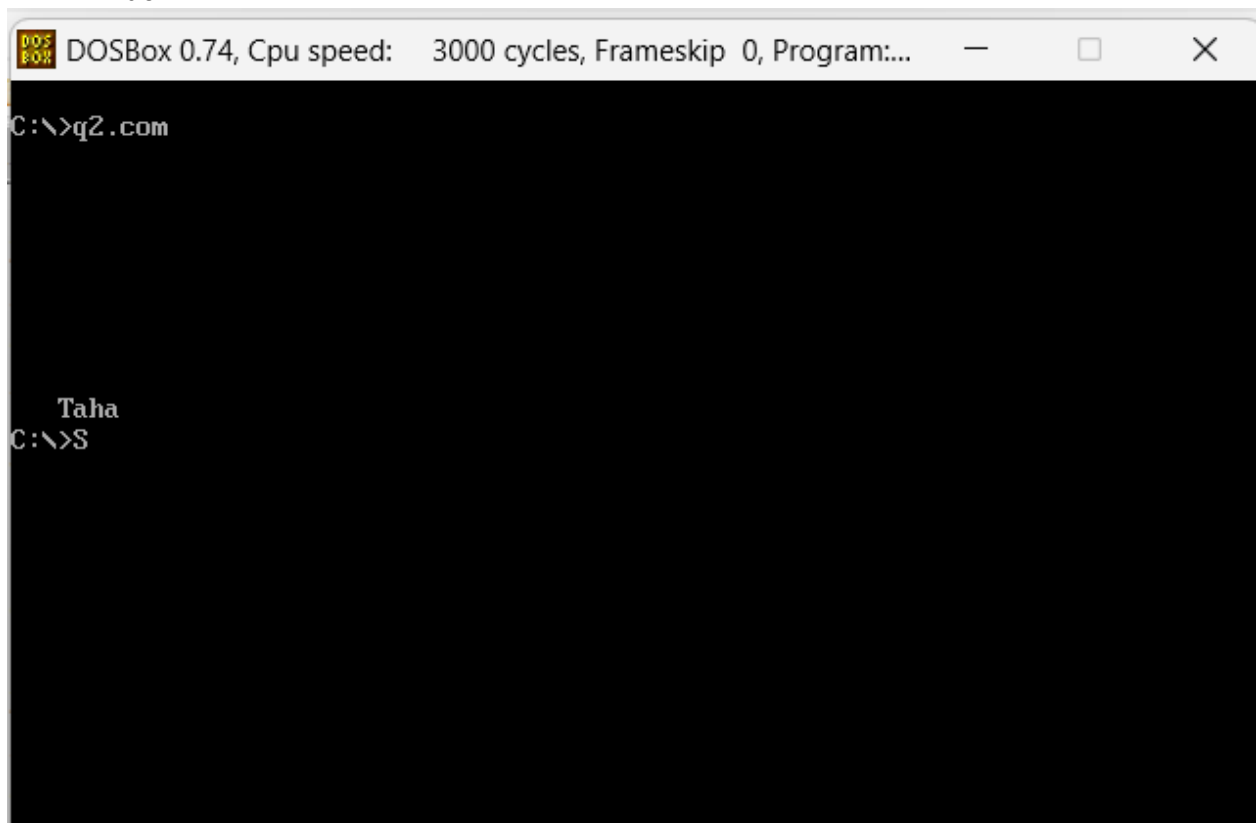
pop es ; segment of string

mov bp, message ; offset of string

int 0x10 ; call BIOS video service

mov ax, 0x4c00 ; terminate program

int 0x21



**Best of
luck**